# FAIR Alignment and Research Infrastructure as Code

## FAIR Alignment and Research Infrastructure as Code I

The template's design aligns with both the original FAIR principles (Wilkinson et al. 2016) and the FAIR for Research Software (FAIR4RS) principles (Barker et al. 2022) at the repository level. FAIR4RS recognizes that software has requirements distinct from data—executability, composability, and dependency management—and the template addresses each.

**Findability.** Outputs are *Findable* through standardized directory structures, manifest files, and machine-readable metadata embedded in PDFs. Every project's `config.yaml` provides structured metadata (title, authors, DOIs, keywords) in a format parseable by both Pandoc and external indexing services. The `metrics.json` output provides a machine-readable inventory of all generated artifacts, their locations, and their provenance hashes.

**Accessibility.** Outputs are *Accessible* via open-source distribution on GitHub, with metadata embedded in the artifact itself rather than in a separate registry. The steganographic layer embeds provenance information directly in the PDF—including SHA-256 content hashes, build timestamps, and QR-encoded metadata—ensuring accessibility even when the PDF circulates outside the repository.

**Interoperability.** *Interoperability* is achieved through standard formats (PDF, JSON, BibTeX, YAML) and well-defined module APIs that enable cross-project composition. The Pandoc rendering pipeline accepts any Markdown-with-LaTeX input conforming to the template's section numbering conventions, allowing seamless migration of manuscripts from other Pandoc-based workflows.

**Reusability.** *Reusability* is ensured by the Standalone Project Paradigm—any project can be extracted and reused independently—and by the Documentation Duality standard, which satisfies FAIRsoft's inspectability and documentation quality indicators (Garijo et al. 2024). The pipeline's automated testing and coverage enforcement directly operationalize the FAIR4RS executability requirement: software that cannot pass its own test suite cannot produce publishable output.

At a higher level of abstraction, `template/` applies the DevOps principle of *Infrastructure as Code* (IaC) to the research lifecycle. In production software engineering, IaC means that server configuration is version-controlled, automatically provisioned, and independently reproducible (Wilson et al. 2017). `template/` extends this principle to the research manuscript: the document is not authored in a word processor and emailed to collaborators, but *built* from version-controlled Markdown sources, *tested* against formal coverage thresholds, and *deployed* to a provenance-embedded PDF.

Every component of the research pipeline—the test suite, the analysis scripts, the rendering configuration, and the steganographic watermark—is specified in code, committed to git, and reproducible from a clean checkout. This deterministic build property means that any researcher can clone the repository, run `./run.sh --pipeline`, and produce a byte-for-byte identical manuscript (modulo timestamps in the steganographic metadata).

Software Heritage (Di Cosmo et al. 2020) provides persistent SWHIDs (Software Hash Identifiers) for source code snapshots, enabling stable citation of any specific version of `template/` as a discrete software artifact—closing the loop from research infrastructure to citable scientific contribution. Combined with Zenodo DOI registration (supported by `infrastructure.publishing`), this creates a dual-identifier citation chain: SWHID for source provenance, DOI for

publication metadata (Katz et al. 2021).